

Proposal: Bayesian Hierarchical Prediction of Used Car Prices

0. Group Members

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1. Topic and Objective

We choose Option 1 and we plan to conduct a complete Bayesian hierarchical model analysis on a real world dataset using Markov chain Monte Carlo (MCMC). Our project will develop and evaluate a Bayesian hierarchical model for predicting used car prices using a publicly available dataset obtained from Kaggle.

The objective of this project is to assess how a Bayesian hierarchical model can predict used car prices from observable vehicle characteristics. We will use variables such as vehicle age, mileage, accident history, transmission type, fuel type, and brand, etc. to predict price. Because vehicle prices may differ across brands so we will incorporate a hierarchical structure that allows baseline prices to vary by brand. This may improve predictive performance by borrowing strength across groups.

2. Data Description

The dataset used in this project is the Used Car Price Prediction Dataset, obtained from Kaggle and created by Taeef Najib:

<https://www.kaggle.com/datasets/taeefnajib/used-car-price-prediction-dataset>

The dataset contains **4,009 observations** of used vehicle listings and includes variables such as price, mileage, model year, fuel type, transmission, accident history, clean title status, and brand/model information.

The main variables relevant to our analysis are:

- Price: listing price of the vehicle (response variable)
- Model Year: used to construct vehicle age
- Mileage: total miles driven
- Fuel Type: type of fuel (gasoline, diesel, electric, hybrid, etc.)
- Transmission: transmission type
- Accident History: whether the vehicle has prior damage
- Clean Title: whether the vehicle has a clean title
- Brand/Model: manufacturer information

We will clean the data by handling missing values and transforming skewed variables when needed. And then we will also create some new variables before fitting the model. For example, we can apply log transformations to variables like price or mileage to improve model fit and predictive performance.

3. Model Description

We will use Bayesian hierarchical regression model to predict used car prices. The response variable is vehicle price and it may be log transformed in order to better satisfy modeling assumptions. Key predictors will include main variables that mentioned above.

we allow the intercept to vary across vehicle brands in order to introduce the hierarchical structure, which means that different brands may have different baseline price levels. This accounts for systematic brand level variation while partially pooling information across brands.

Conditional on the predictors and brand specific intercepts, the response variable will be assumed to follow a normal distribution. Regression coefficients will be assigned normal priors, the brand level intercepts will be modeled through a common group-level distribution and variance parameters will be assigned weakly informative priors. The model will be estimated using MCMC methods implemented in JAGS.

Our main interest is predictive performance so the fitted model will be used to generate posterior predictive distributions for vehicle prices.

4. Analysis Plan

First, we will preprocess the data by handling missing values, selecting relevant variables and constructing transformed variables such as vehicle age and possibly log price or log mileage. We will also split the dataset into training and testing sets so that model performance can be evaluated on held out data.

Second, we will estimate the Bayesian hierarchical regression model in JAGS using the training data and run multiple MCMC chains. We will assess convergence using trace plots and other standard diagnostics.

Third, we will evaluate the predictive performance of the model. We will generate predicted vehicle prices for the test set and assess prediction accuracy using measures by using the posterior predictive distribution, such as root mean squared error (RMSE) and mean absolute error (MAE).

Finally, we will compare the predictive performance of the Bayesian hierarchical model with a standard OLS regression model. We will discuss whether the hierarchical structure improves out-of-sample prediction and summarize the practical strengths and limitations of the model.

5. References

Naji, T. (2023). *Used Car Price Prediction Dataset*. Kaggle.
<https://www.kaggle.com/datasets/taefnajib/used-car-price-prediction-dataset>

Use of Generative AI

Generative AI tools were used to assist with brainstorming. All content was reviewed and verified by the authors.